QIWEI (KEVIN) MA

(+852) 70115761 ◇ mqiwei@eduhk.hk
qiwei-ma.github.io ◇ Hong Kong, China

EDUCATION

Shenzhen University (SZU) *June 2026*
M.S. in Artificial Intelligence
GPA: 3.5/4.0

Ningxia University (NXU) *June 2022*
B.E. in Materials
GPA: 3.0/4.0

INTERESTS

- Agentic AI
- Embodied Intelligence
- AI in Education

PUBLICATIONS

[1] Jun Zhang, **Qiwei Ma**, Yu Zhang, Liying Zhu, and Xiaoming Cao. (2027). From textual lists to in-situ annotations: Empowering junior high school EFL writing with multimodal large language models. *Assessing Writing*, 71, 101146. [DOI].

[2] Juan Yang, Minjuan Wang, Xu Du, and **Qiwei Ma**. (2026). Multimodal affective computing in the era of LLM-based educational agents: Key insights, challenges, and future prospects. *IEEE Transactions on Learning Technologies*, 19, 988–1004. [DOI].

[3] Minjuan Wang, Yizhi Wang, and **Qiwei Ma***. (2026). Agentic AI unleashed: Power, promise, and ethical frontiers. In K. T. Chui, C. Li, J. Jia, S. K. S. Cheung, F. W.-F. Lee, and J. Shang (Eds.), *Blended learning: Innovations for future education* (pp. 27–39). Springer. [DOI].

[4] Jun Zhang, **Qiwei Ma**, Yu Zhang, and Xiaoming Cao. (2026). Multi-agent vs. single-agent AI for EFL speaking practice: A controlled experiment with hybrid input, contextual dialogue, and proficiency-adaptive feedback. *Educational Technology & Society*, 29(2), 297–322. [DOI].

* Corresponding author.

MANUSCRIPTS

Yiqian Xie, Minjuan Wang, Xu Du, **Qiwei Ma**, and Bing Ma. (2026). Toward computational understanding of epistemic emotions in authentic collaboration: A context-aware multimodal benchmark. *IEEE Transactions on Learning Technologies*. Under review.

RESEARCH EXPERIENCE

Multi-Agent Conversational AI for EFL Speaking Practice [4] Jan 2025 - Jan 2026
Supervisor: Dr. Zhang SZU

- Built a multi-agent system for EFL speaking practice with a digital-human interface for contextual dialogue.
- Implemented seven specialised agents for input processing and dialogue supervision, keeping their roles modular so feedback strategies could be revised independently.

- Supported mixed Chinese and English input, with dialogue and feedback adapted to the learner's proficiency.
- Ran a four-week controlled study comparing multi-agent and single-agent versions with university EFL learners.
- Analysed dialogue logs and learner feedback alongside speaking assessments to examine how students used the system and where the two versions differed.

Intelligent Annotation and Feedback System for English Writing [1] Apr 2025 - Sep 2025
Supervisor: Dr. Zhang SZU

- Built MLLM-IAF, a mobile application that assesses handwritten English essays and provides visual feedback for junior high school students.
- Combined multimodal semantic evaluation with OCR to locate errors and place corrections beside the relevant text in an essay.
- Added a teacher review interface for adjusting scores and editing AI-generated feedback.
- Ran a classroom study across four weekly writing tasks, comparing MLLM-IAF with traditional instruction and examining writing performance and learner engagement.
- Checked agreement between AI and teacher scores, and collected student feedback on the system's usefulness and ease of use.

ADDITIONAL RESEARCH

Reasoning for Table Manipulation Mar 2025 - Jan 2026
Supervisors: Prof. Yang, Dr. Tan SIAT, CAS

- Built an LLM framework that uses structured reasoning to carry out table manipulation tasks.
- Constructed a benchmark for restructuring tables, including wide-to-long conversion and parsing fields in semi-structured data.
- Trained models on reasoning traces through supervised fine-tuning followed by GRPO optimisation, then evaluated them across the benchmark tasks.
- Analysed errors at row and column level to understand how missing fields affected the structural integrity of the resulting tables.

Differential Privacy Image Generation Mar 2024 - Apr 2025
Supervisor: Dr. Zhang SZU

- Built a differentially private image generation framework to investigate gradient clipping bias and training stability.
- Added reconstruction loss and noise injection during generator upsampling to improve the utility and diversity of generated images.
- Implemented classifier and encoder components alongside adversarial training, with gradient sanitisation in the training pipeline.
- Evaluated generated images on MNIST and Fashion-MNIST, checking image quality and their usefulness for downstream classification.

EXPERIENCE

Research Assistant Apr 2026 - Present
The Education University of Hong Kong Hong Kong, China

Research Intern May 2025 - Jan 2026
Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences China